

Normal-theory weighting in two-stage estimation: a special case, not a default

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Question and answer

Two-stage estimation of a structural model under missing data (Savalei and Bentler’s ML2S [1]) runs in two steps. Stage 1 fits the saturated model to incomplete data by FIML, returning $\hat{\eta}_S = (\hat{\mu}, \text{vech } \hat{\Sigma})$ and its asymptotic covariance Γ_{FIML} . Stage 2 fits the structural model to $\hat{\eta}_S$ by *normal-theory* maximum likelihood, then corrects the standard errors and the test statistic with Γ_{FIML} . The Stage-2 *weight* is therefore the normal-theory weight $\Gamma_{\text{NT}}(\Sigma)^{-1}$, which is blind to the missingness pattern.

The ordinal (polychoric) pipeline made the opposite choice: given the Stage-1 polychoric correlations, Stage 2 weights by (the diagonal of) the estimated asymptotic covariance of those correlations (DWLS), never by a normal-theory weight. We argue the two conventions answer the *same* question with different defaults. The Stage-2 weight should approximate $\Gamma_{\text{true}}^{-1}$ (or its diagonal); normal-theory weighting is licensed only when $\Gamma_{\text{true}} \approx \Gamma_{\text{NT}}(\Sigma)$, which holds for complete near-normal data and fails, through an external nuisance, both for polychorics (thresholds and cell sparsity) and for FIML under non-trivial missingness (the missingness pattern). The difference between the two cases is only how far Γ_{true} departs from Γ_{NT} , which sets how much an empirical weight can buy.

1 One estimator family, one knob

Write the Stage-2 residual $r(\theta) = \hat{\eta}_S - \eta(\theta)$ and the moment Jacobian $\Delta = \partial\eta/\partial\theta^\top$. Every member of the least-squares family minimizes a quadratic form in r ,

$$F(\theta) = \frac{1}{2} r(\theta)^\top W r(\theta), \quad (1)$$

differing only in the weight W :

ULS	$W = I$	identity
GLS / NT	$W = \Gamma_{\text{NT}}(\Sigma)^{-1}$	$\Gamma_{\text{NT}} = \text{blockdiag}(\Sigma, 2D^+(\Sigma \otimes \Sigma)D^{+\top})$
WLS / ADF	$W = \hat{\Gamma}^{-1}$	full empirical
DWLS	$W = \text{diag}(\hat{\Gamma})^{-1}$	empirical, marginals only
DLS	$W = ((1-a)\Gamma_{\text{NT}} + a\hat{\Gamma})^{-1}$	Browne mix [3]

For a just-identified Stage 1 every such W gives a consistent $\hat{\theta}$. The estimator's asymptotic law is the sandwich

$$\sqrt{n}(\hat{\theta} - \theta_0) \rightsquigarrow \mathcal{N}(0, A^{-1}BA^{-1}), \quad A = \Delta^\top W \Delta, \quad B = \Delta^\top W \Gamma W \Delta, \quad (2)$$

with Γ the true asymptotic covariance of $\sqrt{n} \hat{\eta}_S$ (here Γ_{FIML}). The overall fit statistic is $T = 2n F(\hat{\theta})$, whose limit is $\sum_j \lambda_j \chi_1^2$ with λ_j the nonzero eigenvalues of $U\Gamma$, $U = W - W\Delta(\Delta^\top W \Delta)^{-1}\Delta^\top W$; the Satorra–Bentler scaled statistic divides T by $\bar{c} = \text{tr}(U\Gamma)/\text{df}$ [5].

Remark. Validity of the corrected SE and test is restored by the sandwich and the $U\Gamma$ spectrum for *any* consistent W . The weight is therefore an efficiency and stability knob, not a validity one. That ULS plus the robust correction is competitive with DWLS in ordinal models [6, 7] is the same statement: once the test is corrected, the weight is second order for validity.

Proposition 1. *Within (1), $W = \Gamma^{-1}$ is asymptotically efficient, and then (2) collapses to $A^{-1} = (\Delta^\top \Gamma^{-1} \Delta)^{-1}$ and $U\Gamma$ is idempotent, so $\lambda_j \equiv 1$ and $\bar{c} = 1$. The normal-theory weight $W = \Gamma_{\text{NT}}(\Sigma)^{-1}$ equals Γ^{-1} , and is thus efficient, if and only if $\Gamma = \Gamma_{\text{NT}}(\Sigma)$.*

2 When is $\Gamma \approx \Gamma_{\text{NT}}$?

The normal-theory weight uses the closed form $\Gamma_{\text{NT}}(\Sigma)$, a function of the model-implied second moments alone. It is the correct Γ in exactly one regime.

Proposition 2. *For complete data from a distribution with vanishing excess fourth cumulants, $\Gamma = \Gamma_{\text{NT}}(\Sigma)$. Two departures break this through a nuisance that $\Gamma_{\text{NT}}(\Sigma)$ cannot see:*

- (i) Missingness. *For the saturated FIML estimator under a missing pattern, the precision of each $\hat{\sigma}_{ij}$ depends on the cases that jointly observe*

(i, j) ; Γ_{FIML} is governed by the pattern, not by Σ . At zero missingness $\Gamma_{\text{FIML}} = \Gamma_{\text{NT}}(\Sigma)$ exactly.

- (ii) Categorization. For polychoric correlations the precision depends on thresholds and cell probabilities; Γ_{poly} has no closed form in the model-implied correlation matrix, and $\Gamma_{\text{NT}}(R)$ is never the right weight, at any n .

The asymmetry between (i) and (ii) is one of degree, and it is the practically important part.

Remark (shape versus scale). Under (i) with data missing completely at random, Γ_{FIML} is $\Gamma_{\text{NT}}(\Sigma)$ with heterogeneous per-element effective- n rescaling: the off-diagonal (fourth-moment) structure of Γ_{FIML} is still normal-theory shaped, only the per-element scales are wrong. The normal-theory weight then has the right *shape* and the wrong *scale*, and it degrades gracefully from an exact baseline as missingness grows. Under (ii) $\Gamma_{\text{NT}}(R)$ has both the wrong shape and the wrong scale. Hence DWLS, which fixes the marginal scales but discards the off-diagonals, gains less over NT in the missing-data case than in the ordinal case, and is not a guaranteed improvement there; its advantage grows precisely where Γ_{FIML} stops being a smooth rescaling of Γ_{NT} , namely heavy or strongly pattern-dependent missingness.

3 The transplanted argument

Proposition 2 licenses, by the same reasoning that retired NT weighting for polychorics, the use of a missingness-aware Stage-2 weight in two-stage FIML:

$$\text{DWLS: } W = \text{diag}(\Gamma_{\text{FIML}})^{-1}, \quad \text{ADF: } W = \Gamma_{\text{FIML}}^{-1}, \quad \text{DLS: } W = ((1-a)\Gamma_{\text{NT}} + a\Gamma_{\text{FIML}})^{-1}.$$

Γ_{FIML} is already computed for the robust correction [1], so the additional cost is negligible next to the Stage-1 EM: DWLS is a diagonal rescale, ADF one moment-space inverse, DLS one mix. The mixing scalar a may be fixed or chosen empirically from the Γ_{FIML} -versus- Γ_{NT} departure [8]; the empirical-Bayes choice under missingness needs casewise saturated-score moments and is left open. By Proposition 1 ADF yields $\bar{c} = 1$ and DLS interpolates, $a = 0$ recovering NT and $a = 1$ recovering ADF.

Corollary 1. *The Stage-2 weight changes the point estimate and its efficiency; it does not change the asymptotic validity of the corrected test, which*

the UT spectrum delivers for every consistent W . A simulation should therefore compare (a) finite-sample efficiency of $\hat{\theta}$ and (b) finite-sample calibration of the scaled and eigenvalue (FMG/pEBA) statistics, across $\{NT, DWLS, ADF, DLS\}$, with the contrast expected to be null under light MCAR and to favor the empirical weights only under heavy MAR with non-normal margins.

Remark (implementation). In `magmaan` the four weights are one builder (`two_stage_stage2_weight`, `TwoStageWeight` $\in \{Nt, Dwls, Adf, Dls\}$) over the $[\mu; \text{vech } \Sigma]$ block layout, built from Γ_{NT} and Γ_{FIML} . The robust surface is unchanged: the same Satorra–Bentler eigen-cores and FMG tail transforms consume the UT spectrum produced by whichever W is supplied, so a non-NT Stage-2 fit is a weight swap, not new test machinery. The NT member reproduces `lavaan`’s `robust.two.stage` bit-for-bit.

References

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